

BABU BANARASI DAS **UNIVERSITY**

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SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On
Student Management System Using MongoDB

Submitted to:

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1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Students database.

```
MongoDB Shell — Students

> MONGOSH
use Students
switched to db Students
```

2. CREATE COLLECTION

MongoDB Query:

db.createCollection("students")

Explanation: Creates a collection named students.

```
MongoDB Shell — Students

> MONGOSH
db.createCollection("students")
{ ok: 1 }
```

3. INSERT 50 STUDENTS

MongoDB Query:

db.students.insertMany([...])

Explanation: Inserts 50 student documents into the collection.

```
MongoDB Shell — Students

> MONGOSH
db.students.insertMany([
  { _id: ObjectId('6f1a2e9c78f0f98eb52df200'), roll: 1, name: 'Ritika', age: 18, marks: 51, city:
'Ayodhya' },
  { _id: ObjectId('6f1a2e9c78f0f98eb52df201'), roll: 2, name: 'Devansh', age: 19, marks: 64, city:
'Varanasi' },
  { _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city:
'Gonda' },
  // ... 47 more student documents
])
{ acknowledged: true, insertedCount: 50 }
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

```
db.students.find()
```

Explanation: Retrieves all documents from the students collection.

```
MongoDB Shell — Students

> MONGOSH
db.students.find()
{ _id: ObjectId('6f1a2e9c78f0f98eb52df200'), roll: 1, name: 'Ritika', age: 18, marks: 51, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df201'), roll: 2, name: 'Devansh', age: 19, marks: 64, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df203'), roll: 4, name: 'Yash', age: 18, marks: 87, city: 'Jaunpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 60, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df205'), roll: 6, name: 'Aarav', age: 19, marks: 64, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df206'), roll: 7, name: 'Simran', age: 18, marks: 85, city: 'Prayagraj' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df207'), roll: 8, name: 'Shivam', age: 21, marks: 88, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df208'), roll: 9, name: 'Ishita', age: 20, marks: 88, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df209'), roll: 10, name: 'Rohit', age: 21, marks: 55, city: 'Ayodhya' }
Type "it" for more (40 more documents)
```

5. EQUAL TO --- \$eq

MongoDB Query:

```
db.students.find({ marks: { $eq: 88 } })
```

Explanation: Finds students whose marks are exactly 88.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ marks: { $eq: 88 } })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df207'), roll: 8, name: 'Shivam', age: 21, marks: 88, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df208'), roll: 9, name: 'Ishita', age: 20, marks: 88, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df217'), roll: 24, name: 'Karan', age: 18, marks: 88, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df227'), roll: 40, name: 'Gaurav', age: 21, marks: 88, city: 'Kanpur' }
```

6. GREATER THAN --- \$gt

MongoDB Query:

```
db.students.find({ marks: { $gt: 85 } })
```

Explanation: Finds students whose marks are greater than 85.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ marks: { $gt: 85 } })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df203'), roll: 4, name: 'Yash', age: 18, marks: 87, city: 'Jaunpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df207'), roll: 8, name: 'Shivam', age: 21, marks: 88, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df208'), roll: 9, name: 'Ishita', age: 20, marks: 88, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20f'), roll: 16, name: 'Rajat', age: 20, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df210'), roll: 17, name: 'Swati', age: 20, marks: 86, city: 'Prayagraj' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df214'), roll: 21, name: 'Niharika', age: 21, marks: 90, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df216'), roll: 23, name: 'Radhika', age: 19, marks: 92, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df217'), roll: 24, name: 'Karan', age: 18, marks: 88, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21a'), roll: 27, name: 'Anjali', age: 19, marks: 93, city: 'Gorakhpur' }
(8 more documents)
```

7. LESS THAN --- \$lt

MongoDB Query:

```
db.students.find({ marks: { $lt: 60 } })
```

Explanation: Finds students whose marks are less than 60.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ marks: { $lt: 60 } })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df200'), roll: 1, name: 'Ritika', age: 18, marks: 51, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df209'), roll: 10, name: 'Rohit', age: 21, marks: 55, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20b'), roll: 12, name: 'Aryan', age: 18, marks: 55, city: 'Jaunpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20d'), roll: 14, name: 'Kunal', age: 20, marks: 52, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df211'), roll: 18, name: 'Manish', age: 18, marks: 52, city: 'Prayagraj' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df212'), roll: 19, name: 'Pooja', age: 20, marks: 55, city: 'Prayagraj' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df225'), roll: 38, name: 'Harshit', age: 21, marks: 55, city: 'Lucknow' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df226'), roll: 39, name: 'Riya', age: 18, marks: 59, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df230'), roll: 49, name: 'Kabir', age: 19, marks: 59, city: 'Gorakhpur' }
```

8. GREATER THAN OR EQUAL TO --- \$gte

MongoDB Query:

```
db.students.find({ marks: { $gte: 90 } })
```

Explanation: Finds students whose marks are 90 or above.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ marks: { $gte: 90 } })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20f'), roll: 16, name: 'Rajat', age: 20, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df214'), roll: 21, name: 'Niharika', age: 21, marks: 90, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df216'), roll: 23, name: 'Radhika', age: 19, marks: 92, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21a'), roll: 27, name: 'Anjali', age: 19, marks: 93, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21f'), roll: 32, name: 'Nikhil', age: 21, marks: 91, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df221'), roll: 34, name: 'Arjun', age: 19, marks: 97, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df222'), roll: 35, name: 'Tanvi', age: 20, marks: 97, city: 'Agra' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22a'), roll: 43, name: 'Rohan', age: 18, marks: 93, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22b'), roll: 44, name: 'Aditi', age: 20, marks: 99, city: 'Gorakhpur' }
(1 more documents)
```

9. LESS THAN OR EQUAL TO --- \$lte

MongoDB Query:

```
db.students.find({ marks: { $lte: 70 } })
```

Explanation: Finds students whose marks are 70 or below.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ marks: { $lte: 70 } })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df200'), roll: 1, name: 'Ritika', age: 18, marks: 51, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df201'), roll: 2, name: 'Devansh', age: 19, marks: 64, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 60, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df205'), roll: 6, name: 'Aarav', age: 19, marks: 64, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df209'), roll: 10, name: 'Rohit', age: 21, marks: 55, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20a'), roll: 11, name: 'Mahi', age: 19, marks: 63, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20b'), roll: 12, name: 'Aryan', age: 18, marks: 55, city: 'Jaunpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20d'), roll: 14, name: 'Kunal', age: 20, marks: 52, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df211'), roll: 18, name: 'Manish', age: 18, marks: 52, city: 'Prayagraj' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df212'), roll: 19, name: 'Pooja', age: 20, marks: 55, city: 'Prayagraj' }
(11 more documents)
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(
  { roll: 5 },
  { $set: { marks: 95 } }
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.

```
MongoDB Shell — Students

> MONGOSH
db.students.updateOne(
  { roll: 5 },
  { $set: { marks: 95 } }
)
{ acknowledged: true, insertedId: null, matchedCount: 1, modifiedCount: 1, upsertedCount: 0 }

// updated document:
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 95, city: 'Kanpur' }
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.students.deleteOne({ roll: 10 })
```

Explanation: Deletes one document matching roll number 10.

```
MongoDB Shell — Students

> MONGOSH
db.students.deleteOne({ roll: 10 })
{ acknowledged: true, deletedCount: 1 }
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({
  marks: { $lt: 60 }
})
```

Explanation: Deletes all students having marks below 60.

```
MongoDB Shell — Students

> MONGOSH
db.students.deleteMany({
  marks: { $lt: 60 }
})
{ acknowledged: true, deletedCount: 8 }
```

13. AND OPERATION --- \$and

MongoDB Query:

```
db.students.find({
  $and: [
    { marks: { $gt: 80 } },
    { age: 19 }
  ]
})
```

Explanation: Finds students satisfying both conditions.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({
  $and: [
    { marks: { $gt: 80 } },
    { age: 19 }
  ]
})
{ _id: ObjectId('6f1a2e9c78f0f98eb52df216'), roll: 23, name: 'Radhika', age: 19, marks: 92, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21a'), roll: 27, name: 'Anjali', age: 19, marks: 93, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df221'), roll: 34, name: 'Arjun', age: 19, marks: 97, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df231'), roll: 50, name: 'Sanya', age: 19, marks: 84, city: 'Gonda' }
```

14. OR OPERATION --- \$or

MongoDB Query:

```
db.students.find({
  $or: [
    { city: "Lucknow" },
    { city: "Kanpur" }
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.

```
MongoDB Shell — Students

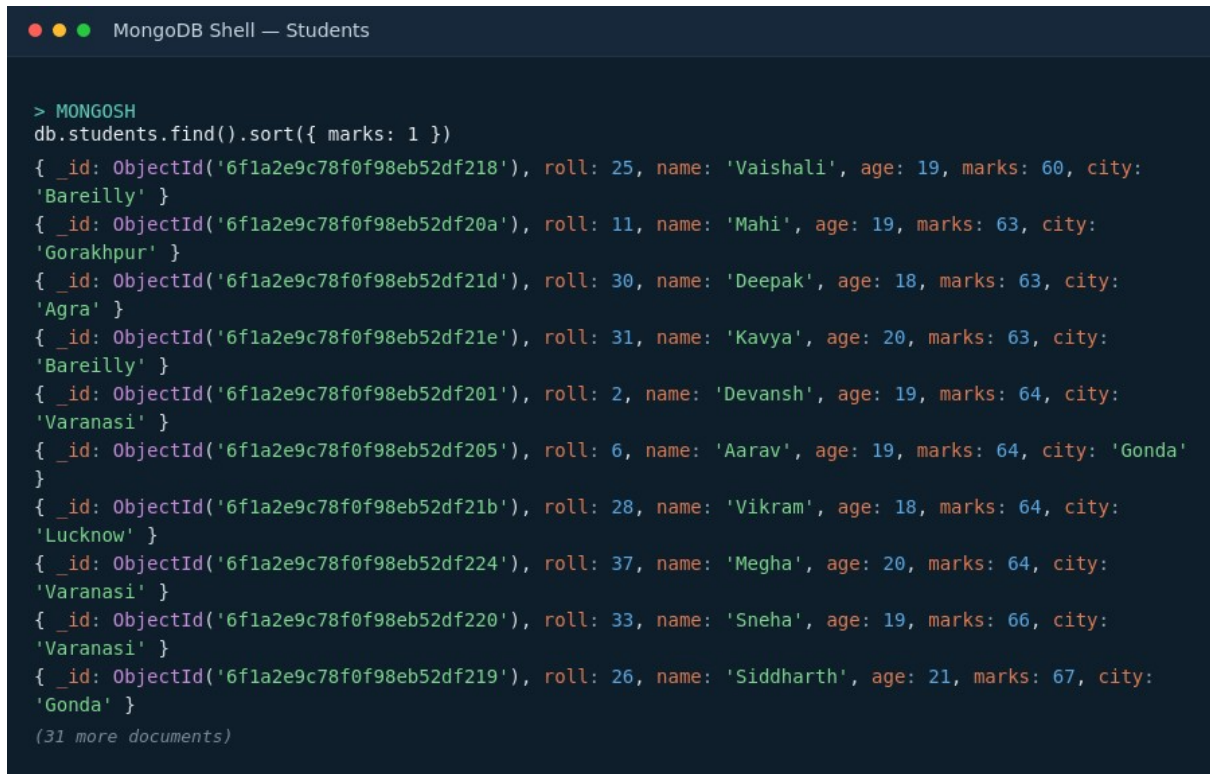
> MONGOSH
db.students.find({
  $or: [
    { city: "Lucknow" },
    { city: "Kanpur" }
  ]
})
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 95, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20e'), roll: 15, name: 'Diksha', age: 18, marks: 74, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21b'), roll: 28, name: 'Vikram', age: 18, marks: 64, city: 'Lucknow' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df227'), roll: 40, name: 'Gaurav', age: 21, marks: 88, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22a'), roll: 43, name: 'Rohan', age: 18, marks: 93, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22d'), roll: 46, name: 'Charu', age: 19, marks: 79, city: 'Lucknow' }
```

15. SORT OPERATION (Ascending)

MongoDB Query:

```
db.students.find().sort({ marks: 1 })
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.



```
MongoDB Shell — Students

> MONGOSH
db.students.find().sort({ marks: 1 })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df218'), roll: 25, name: 'Vaishali', age: 19, marks: 60, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20a'), roll: 11, name: 'Mahi', age: 19, marks: 63, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21d'), roll: 30, name: 'Deepak', age: 18, marks: 63, city: 'Agra' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21e'), roll: 31, name: 'Kavya', age: 20, marks: 63, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df201'), roll: 2, name: 'Devansh', age: 19, marks: 64, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df205'), roll: 6, name: 'Aarav', age: 19, marks: 64, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21b'), roll: 28, name: 'Vikram', age: 18, marks: 64, city: 'Lucknow' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df224'), roll: 37, name: 'Megha', age: 20, marks: 64, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df220'), roll: 33, name: 'Sneha', age: 19, marks: 66, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df219'), roll: 26, name: 'Siddharth', age: 21, marks: 67, city: 'Gonda' }
(31 more documents)
```

15. SORT OPERATION (Descending)

MongoDB Query:

```
db.students.find().sort({ marks: -1 })
```

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
MongoDB Shell — Students

> MONGOSH
db.students.find().sort({ marks: -1 })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22b'), roll: 44, name: 'Aditi', age: 20, marks: 99, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df221'), roll: 34, name: 'Arjun', age: 19, marks: 97, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df222'), roll: 35, name: 'Tanvi', age: 20, marks: 97, city: 'Agra' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 95, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21a'), roll: 27, name: 'Anjali', age: 19, marks: 93, city: 'Gorakhpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22a'), roll: 43, name: 'Rohan', age: 18, marks: 93, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df216'), roll: 23, name: 'Radhika', age: 19, marks: 92, city: 'Ayodhya' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21f'), roll: 32, name: 'Nikhil', age: 21, marks: 91, city: 'Bareilly' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df20f'), roll: 16, name: 'Rajat', age: 20, marks: 90, city: 'Agra' }
(31 more documents)
```

16. LIMIT OPERATION

MongoDB Query:

```
db.students.find().limit(5)
```

Explanation: Displays only the first five documents.

```
MongoDB Shell — Students

> MONGOSH
db.students.find().limit(5)
{ _id: ObjectId('6f1a2e9c78f0f98eb52df201'), roll: 2, name: 'Devansh', age: 19, marks: 64, city: 'Varanasi' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df202'), roll: 3, name: 'Tanya', age: 18, marks: 93, city: 'Gonda' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df203'), roll: 4, name: 'Yash', age: 18, marks: 87, city: 'Jaunpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df204'), roll: 5, name: 'Kritika', age: 18, marks: 95, city: 'Kanpur' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df205'), roll: 6, name: 'Aarav', age: 19, marks: 64, city: 'Gonda' }
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({ city: "Lucknow" })
```

Explanation: Finds all students whose city is Lucknow.

```
MongoDB Shell — Students

> MONGOSH
db.students.find({ city: "Lucknow" })
{ _id: ObjectId('6f1a2e9c78f0f98eb52df21b'), roll: 28, name: 'Vikram', age: 18, marks: 64, city: 'Lucknow' }
{ _id: ObjectId('6f1a2e9c78f0f98eb52df22d'), roll: 46, name: 'Charu', age: 19, marks: 79, city: 'Lucknow' }
```

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts documents currently present in the collection (performed after the delete operations).

```
MongoDB Shell — Students

> MONGOSH
db.students.countDocuments()
41
```